

Naive Bayes Multiclass Document Classification

From Word Counts to Probabilistic Classification

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1. Document Classification

Document Classification

Document classification assigns one or more predefined labels to a document based on its textual content.

Examples for Arabic news:

- Sports
- Politics
- Economy
- Technology
- Culture
- International

Goal

Given a document d , predict its class:

$$\hat{c} = \arg \max_{c \in C} P(c \mid d)$$

2. From Text to Naive Bayes



Multinomial Naive Bayes

The classifier uses the frequencies of words in a document and estimates how likely those words are under each class.

3. Bayes' Rule

Bayes' theorem relates a posterior probability to a prior and a likelihood:

$$P(c \mid d) = \frac{P(d \mid c)P(c)}{P(d)}$$

where:

- $P(c \mid d)$: probability of class c given document d
- $P(d \mid c)$: probability of observing document d in class c
- $P(c)$: prior probability of class c
- $P(d)$: probability of the document

For classification, $P(d)$ is the same for all classes.

Therefore:

$$P(c \mid d) \propto P(d \mid c)P(c)$$

4. Naive Bayes Assumption

Let:

$$d = (w_1, w_2, \dots, w_n)$$

be a document containing n words.

Naive Bayes assumes conditional independence between features given the class:

$$P(d \mid c) = \prod_{i=1}^n P(w_i \mid c)$$

Therefore:

$$P(c \mid d) \propto P(c) \prod_{i=1}^n P(w_i \mid c)$$

Why “Naive”?

The words in a document are not truly independent. The independence assumption is a simplifying approximation that makes the model computationally efficient.

5. Multinomial Naive Bayes

For text classification, the **Multinomial Naive Bayes** model uses word occurrence counts.

Let: n_{wc} = number of occurrences of word w in class c and:

$$N_c = \sum_{w \in V} n_{wc}$$

be the total number of word occurrences in class c .

Without smoothing:

$$P(w \mid c) = \frac{n_{wc}}{N_c}$$

The document score is:

$$P(c) \prod_{w \in d} P(w \mid c)^{n_{wd}}$$

where n_{wd} is the number of occurrences of w in document d .

6. Estimating the Class Prior

The prior probability of a class is estimated from the training documents.

If:

N = total number of training documents

and:

N_c = number of documents in class c

then:

$$P(c) = \frac{N_c}{N}$$

Example:

Class	Documents	Prior
Sports	400	0.40
Politics	300	0.30
Economy	200	0.20

7. Estimating Word Probabilities

Suppose the vocabulary is:

$$V = \{\text{team, goal, market, stock, match}\}$$

For the Sports class, suppose the word counts are:

Word	team	goal	market	stock	match
Count	20	15	0	0	10

The total number of word occurrences is:

$$N_{\text{Sports}} = 45$$

Without smoothing:

$$P(\text{market} \mid \text{Sports}) = \frac{0}{45} = 0$$

This zero causes an important problem.

8. The Zero-Probability Problem

Suppose a new document contains:

`d = team market match`

For Sports:

$$P(d \mid \text{Sports}) = P(\text{team} \mid S)P(\text{market} \mid S)P(\text{match} \mid S)$$

If:

$$P(\text{market} \mid S) = 0$$

then:

$$P(d \mid S) = 0$$

regardless of the other words.

Problem

One unseen word can force the entire document likelihood to zero.

9. Laplace Smoothing

For a vocabulary of size $|V|$, Laplace smoothing changes:

$$\frac{n_{wc}}{N_c}$$

to:

$$P(w \mid c) = \frac{n_{wc} + 1}{N_c + |V|}$$

The numerator is increased by one for every word.

The denominator must also increase because one count is added to every vocabulary item.

General Add- α Smoothing

$$P(w \mid c) = \frac{n_{wc} + \alpha}{N_c + \alpha|V|}$$

Laplace smoothing corresponds to:

$$\alpha = 1$$

10. Laplace Smoothing: Worked Example

Recall:

$$N_{\text{Sports}} = 45$$

and:

$$|V| = 5$$

For the unseen word market:

$$n_{\text{market},S} = 0$$

After Laplace smoothing:

$$P(\text{market} \mid S) = \frac{0 + 1}{45 + 5} = \frac{1}{50} = 0.02$$

For team:

$$P(\text{team} \mid S) = \frac{20 + 1}{45 + 5} = \frac{21}{50} = 0.42$$

For goal:

11. Why the Denominator Changes

Suppose:

$$V = \{a, b, c, d\}$$

and a class contains:

$$N_c = 10$$

word occurrences.

Before smoothing:

$$P(a \mid c) + P(b \mid c) + P(c \mid c) + P(d \mid c) = 1$$

With Laplace smoothing, one count is added to every word:

$$N_c + |V| = 10 + 4 = 14$$

Therefore:

$$\sum_{w \in V} \frac{n_{wc} + 1}{N_c + |V|} = \frac{N_c + |V|}{N_c + |V|} = 1$$

12. What if a Test Word Is Not in the Vocabulary?

Suppose the training vocabulary is:

$$V = \{\text{team}, \text{goal}, \text{match}\}$$

but the test document contains:

stadium

There are two common strategies.

Strategy 1: Ignore Unknown Words

Discard words that are not in the training vocabulary.

Strategy 2: Use an <UNK> Token

Include an unknown token in the vocabulary during training and map unseen test words to:

<UNK>

Practical point

13. Computing the Multiclass Score

For document d and class c :

$$P(c \mid d) \propto P(c) \prod_{w \in d} P(w \mid c)^{n_{wd}}$$

Direct multiplication can produce extremely small numbers.

Instead, use logarithms:

$$\log P(c \mid d) = \log P(c) + \sum_{w \in d} n_{wd} \log P(w \mid c) + K$$

where K is the same for every class and can be ignored.

Thus:

$$\text{score}(c, d) = \log P(c) + \sum_{w \in d} n_{wd} \log P(w \mid c)$$

14. Worked Multiclass Example

Consider three classes:

$$C = \{\text{Sports, Economy, Politics}\}$$

Suppose:

$$P(S) = 0.5, \quad P(E) = 0.3, \quad P(P) = 0.2$$

For a document:

$$d = \text{team market}$$

the classifier computes:

$$\text{score}(S) = \log P(S) + \log P(\text{team} \mid S) + \log P(\text{market} \mid S)$$

and similarly:

$$\text{score}(E)$$

$$\text{score}(P)$$

15. Multinomial Naive Bayes Algorithm

- 1 Build the vocabulary from the training set.
- 2 Count documents in each class.
- 3 Compute class priors:

$$P(c) = \frac{N_c}{N}$$

- 4 Count word occurrences for every class.
- 5 Apply Laplace smoothing:

$$P(w | c) = \frac{n_{wc} + 1}{N_c + |V|}$$

- 6 For each test document, compute the log score for every class.
- 7 Select the class with the largest score.

Count → Smooth → Log → Score → Predict

16. Core Data Structures

A simple implementation can maintain:

```
class_doc_count = {}  
word_count = {}  
total_word_count = {}
```

For example:

```
class_doc_count["sports"] = 400  
  
word_count["sports"]["goal"] = 120  
word_count["sports"]["team"] = 95  
  
total_word_count["sports"] = 5000
```

The vocabulary is shared:

```
vocabulary = {  
    "goal", "team", "match",  
    "market", "stock", ...  
}
```

17. Training the Classifier

```
from collections import Counter, defaultdict

class_doc_count = Counter()
word_count = defaultdict(Counter)
total_word_count = Counter()
vocabulary = set()

for document, label in train_data:

    class_doc_count[label] += 1

    words = tokenize(document)
    vocabulary.update(words)

    for word in words:
        word_count[label][word] += 1
        total_word_count[word] += 1
```

After processing the training set, the model has the statistics needed to estimate:

$$P(c) \quad \text{and} \quad P(w \mid c)$$

18. Laplace-Smoothing Function

```
def word_probability(word, label):  
    count = word_count[label][word]  
    total = total_word_count[label]  
    V = len(vocabulary)  
  
    return (count + 1) / (total + V)
```

The corresponding mathematical expression is:

$$P(w \mid c) = \frac{n_{wc} + 1}{N_c + |V|}$$

Notice that even when:

$$n_{wc} = 0$$

we obtain:

$$P(w \mid c) > 0$$

19. Prediction Function

```
import math

def predict(document):
    scores = {}

    words = tokenize(document)
    counts = Counter(words)

    N = sum(class_doc_count.values())

    for label in class_doc_count:

        prior = class_doc_count[label] / N
        score = math.log(prior)

        for word, count in counts.items():

            if word not in vocabulary:
                continue

            p = word_probability(word, label)
            score += count * math.log(p)

        scores[label] = score

    return max(scores, key=scores.get)
```

20. Why Use Log Probabilities?

Without logarithms:

$$P(c) \prod_{w \in d} P(w \mid c)^{n_{wd}}$$

may involve multiplying hundreds or thousands of numbers smaller than one.

For example:

$$0.1 \times 0.01 \times 0.001 \times \dots$$

can underflow numerically.

Using:

$$\log(ab) = \log a + \log b$$

gives:

$$\log P(c) + \sum_w n_{wd} \log P(w \mid c)$$

21. Evaluating a Multiclass Classifier

For C classes, construct a confusion matrix:

M_{ij} = number of documents with true class i predicted as j

For example:

	Sports	Economy	Politics
Sports	92	5	3
Economy	7	88	5
Politics	4	6	90

Useful metrics include:

- Accuracy
- Precision
- Recall
- Macro-F1
- Micro-F1

22. Naive Bayes: Advantages and Limitations

Advantages

- Simple probabilistic model.
- Very fast training.
- Fast prediction.
- Works well with sparse text features.
- Small memory footprint.
- Easy to implement and interpret.

Limitations

- Conditional independence assumption is simplistic.
- Ignores word order.
- Vocabulary and preprocessing affect performance.
- Rare words can produce unreliable estimates.
- Does not directly model semantic similarity.

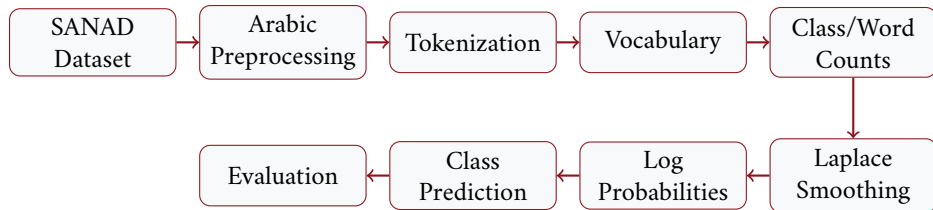
23. Assignment: SANAD Arabic Document Classification

Objective

Implement a **Multinomial Naive Bayes document classifier** for the SANAD Arabic news dataset.

- 1 Load the SANAD training, validation, and test data.
- 2 Apply Arabic text preprocessing.
- 3 Tokenize the documents.
- 4 Build a vocabulary from the training set.
- 5 Count documents and word frequencies for each class.
- 6 Estimate class priors.
- 7 Apply Laplace smoothing to word probabilities.
- 8 Implement classification using log probabilities.
- 9 Evaluate the classifier on the test set.

24. SANAD Assignment: Processing Pipeline



25. SANAD Assignment: Required Experiments

Students should investigate the effect of different feature representations.

Experiment	Representation	Purpose
1	Unigram counts	Basic baseline
2	Bigram counts	Add local context
3	Unigram + bigram	Combined features
4	No smoothing	Observe zero probabilities
5	Laplace smoothing	Handle zero counts
6	TF-IDF + NB	Compare feature weighting

Report

For each experiment, report accuracy, macro-F1, and a confusion matrix, and explain how smoothing and feature choices affect the results.

26. SANAD Assignment: Deliverables

Implementation

Submit a complete Python implementation of Multinomial Naive Bayes.

Required Components

- Data loading
- Arabic preprocessing
- Tokenization
- Vocabulary construction
- Class and word counting
- Laplace smoothing
- Log-probability calculation
- Multiclass prediction
- Evaluation

Analysis

27. Key Takeaways

- 1 Naive Bayes converts word statistics into class probabilities.
- 2 Multinomial Naive Bayes is particularly natural for count-based document features.
- 3 Bayes' rule provides the probabilistic foundation:

$$P(c \mid d) \propto P(c)P(d \mid c)$$

- 4 The naive assumption factorizes the document likelihood.
- 5 Laplace smoothing prevents zero word probabilities:

$$P(w \mid c) = \frac{n_{wc} + 1}{N_c + |V|}$$

- 6 Log probabilities make the implementation numerically stable.
- 7 SANAD provides a realistic Arabic document-classification problem for implementing the complete pipeline.

Any Questions?

Naive Bayes Multiclass Document Classification

Counts $\rightarrow$ Probabilities $\rightarrow$ Laplace Smoothing $\rightarrow$ Classification